POLYDEX PREMIUM

HOT AND COLD WATER SYSTEM

The New
4th Generation
PP-RCT
Pipes & Fittings





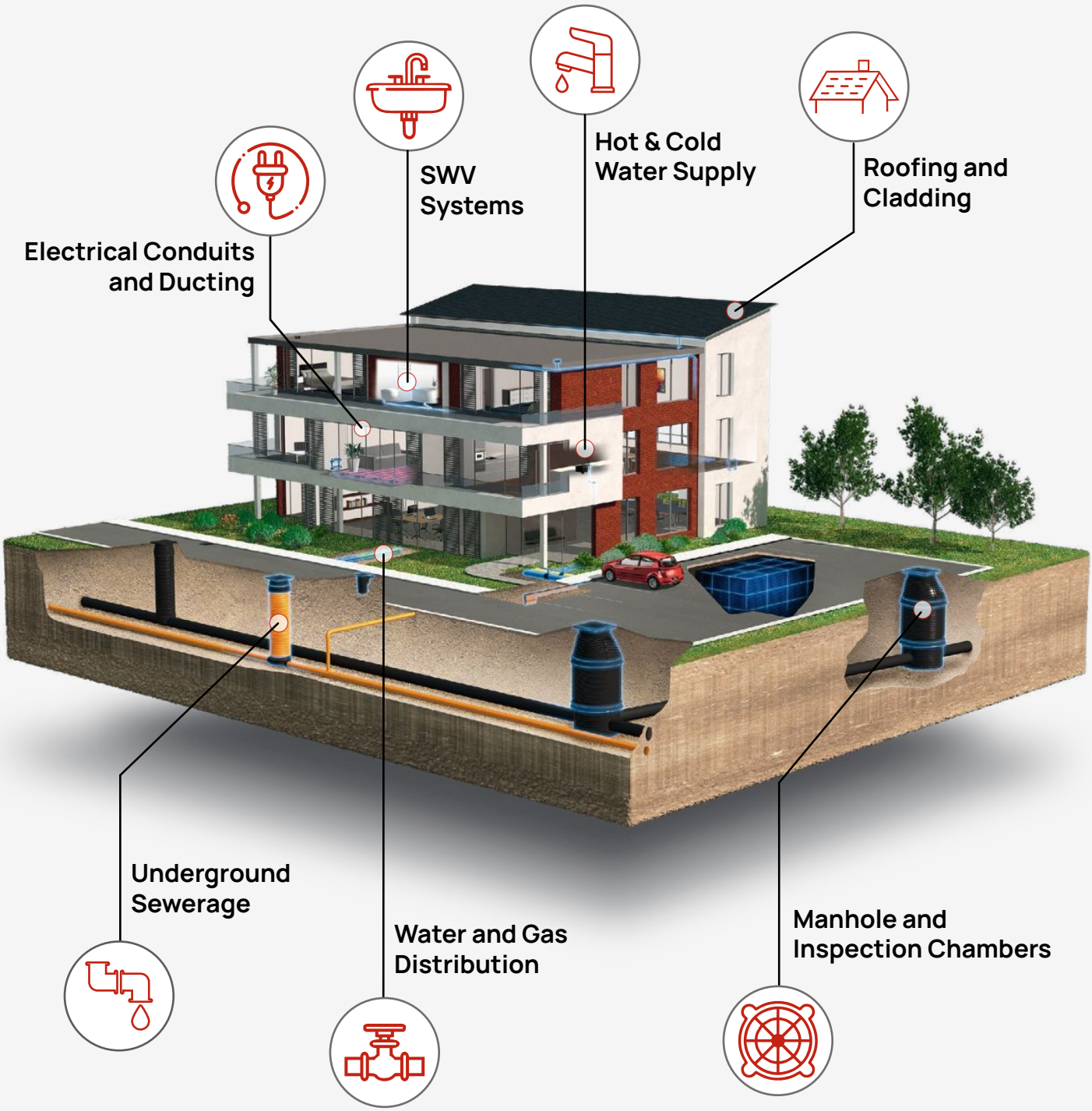
PP-RCT

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DADEX

DADEX

Leader in Roofing and Thermoplastic Piping Systems



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ABOUT DADEX

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facilities in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 66 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

EXTENSIVE PRODUCT RANGE

Water Supply - Infra-Structure

- Aquadex (UPVC Pressure Pipe System)
- T-Flex (PE 100 Pipes & Fittings)

Water Supply - Housing

- Polydex Premium (PP-RCT Pipe System)
- Polydex (PP-R Pipe System)
- Thermoline (PP-R Pipe System)
- Aquadex (UPVC Pressure Pipe System)

Soil, Waste and Vent (SWV)

- Nikasi - Solvent (UPVC Pipe System)
- Nikasi - Rubber Ring (UPVC Pipe System)
- Polyduct (Low Noise System)

Underground Sewerage and Drainage

- Flow Line (UPVC Underground Sewerage System)
- Inspection Chambers and Manholes

Cable Ducting

- PE Cable Duct Pipes
- Electroduct (UPVC Conduits as per IEC - EN Standard)
- Conduits (UPVC Conduits as per Dadex Standard)

Gas Supply

- T-Flex for gas (PE 80 and PE 100)

Introduction

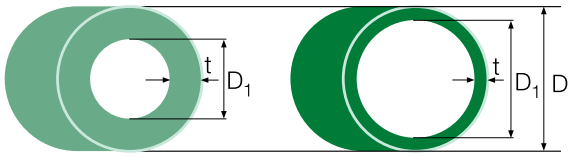
Polydex Premium is designed for transport of hot and cold water. It is also used for the distribution of compressed air. Polydex Premium is the first complete PP piping system in Pakistan that conforms to international quality and standard. In order to take advantage of its chemical resistance and other system properties and use of system for transport of other liquids, gaseous or solid - an individual assessment of any such case would be needed.

Polydex Premium pipe systems are available with high quality PP-RCT pipes and fittings. The PP-RCT is the new 4th generation material have slimmer wall thickness. The symbols used for PP-RCT indicate the fact that these are random copolymers (R=random), with modified crystalline structure (C=crystalline). In addition, PP-RCT is characterized by improved long-term hydrostatic strength, particularly at elevated temperature (T=temperature).

Material

Polydex Premium pipes and fittings are manufactured from PP-RCT material which is 100% certified food-grade material. Its resistance to high temperatures has made Polydex Premium PP-RCT a popular piping system recommended for domestic, commercial buildings and industrial usage. The physical and chemical properties of PP-RCT make it a superior and safe piping system for supply of potable water and other fluids.

Polydex Premium PP-RCT pipes and fittings are supplied in green colour.



Explanatory notes:

- t = wall thickness
- D = external diameter
- D_1 = internal diameter

Standards & Specifications

Polydex Premium PP-RCT pipe system conforms to following international standards:

Pipes	EN ISO 15874 / DIN 8077-8078
Fittings	EN ISO 15874

Complete range of imported PP-RCT fittings is available. Special fittings could be arranged, such as flange connections and short by-pass bends.

Available Range

Polydex Premium PP-RCT piping system is manufactured in the metric sizes of:

20mm, 25mm, 32mm, 40mm, 50mm, 63mm, 75mm, 90mm, 110mm, 125mm, 160mm, 200mm and 250mm.

Applications

Polydex Premium PP-RCT is a plumbing & mechanical hot & cold-water pressure piping distribution system. It is designed for a wide range of applications, such as residential, commercial, HVAC and industrial. Its chemical resistant composition and the high pressure & temperature performance rating, makes the system suitable for a wide range of applications. Some of those most common field of applications are:

- Hot & cold potable water plumbing distribution, for residential and commercial
- Geothermal heating and cooling systems
- Irrigation
- Chilled water and condenser water for cooling towers, data centers, supercomputers
- Reclaimed water collection and distribution
- Industrial and process piping for applications such as chemical processes, high-purity pharmaceuticals, and semiconductor manufacturing.
- Hydronic distribution to radiators, convectors, fan coils, chilled beams, etc.
- Ship building industry



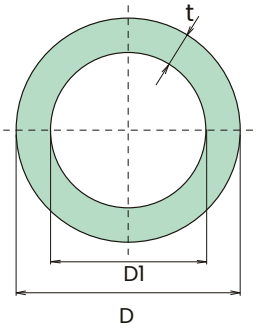
Role of PP-RCT in Sustainability

- Plastic is widely used and preferred, especially in indoor plumbing applications, where it replaces traditional metal pipes.
- Plastic pipes are favored for their lower weight, facilitating easy and quick installation.
- Durability is a key advantage of plastic pipes, contributing to a long service life.
- Plastic pipes prevent the accumulation of dirt and do not have a negative impact on water quality.
- Plastic pipes adhere to strict hygiene standards, making them a reliable choice for plumbing.
- Within the realm of plastic pipes, there is a diverse range of options available, similar to metal pipes.
- PP-RCT pipes, a type of plastic pipe, offer increased pressure resistance at higher temperatures, enhancing safety and durability.
- Due to their properties, PP-RCT pipes can have a reduced wall thickness compared to PP-R pipes for the same applications.
- PP-RCT pipes provide a 37% higher flow rate compared to the previous generation of pipes.

Product Catalog – Pipes

- Polydex Premium PP-RCT Pipes: 20-25 mm, SDR 7.4
- Polydex Premium PP-RCT Pipes: 32-125 mm, SDR 9.0
- Polydex Premium PP-RCT Pipes: 160-250 mm, SDR 11.0

SDR	D mm	D_1 mm	t mm
7.4	20	14.4	2.8
7.4	25	18.0	3.5
9.0	32	24.8	3.6
9.0	40	31.0	4.5
9.0	50	38.8	5.6
9.0	63	48.8	7.1
9.0	75	58.2	8.4
9.0	90	69.8	10.1
9.0	110	85.4	12.3
9.0	125	97.0	14.0
11.0	160	130.8	14.6
11.0	200	163.6	18.2
11.0	250	204.6	22.7





Welding PP-RCT

Table for polyfusion welding

D [mm]	20	25	32	40	50	63	75	90	110	125
Heating time [s]	5	7	8	12	18	24	30	40	50	60

Do not unnecessarily prolong the pipe heating time!

To achieve a proper joint

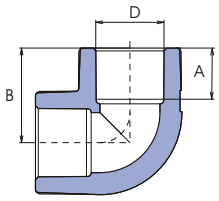
it is necessary not to exceed the appropriate heating times for individual pipe diameters and to set the welding temperature at 260 °C.

Product Catalog – Fittings

Elbow 90 °

SDR	D mm	A mm	B mm
7.4	20	14.5	26.8
7.4	25	16.0	31.0
9.0	32	18.0	36.5
9.0	40	20.5	43.0
9.0	50	23.5	51.0
9.0	63	27.5	60.0
9.0	75	30.0	70.1
9.0	90	33.0	82.0
9.0	110	37.0	95.0
9.0	125	40.0	105.0

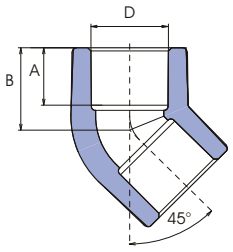




Elbow 45 °

SDR	D mm	A mm	B mm
7.4	20	14.5	20.8
7.4	25	16.0	24.0
9.0	32	18.0	27.0
9.0	40	20.5	31.5
9.0	50	23.5	36.5
9.0	63	27.5	43.0
9.0	75	30.0	48.1
9.0	90	33.0	54.1
9.0	110	37.0	62.1
9.0	125	40.0	69.8

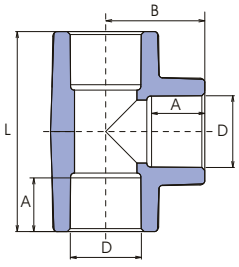




Tee

SDR	D mm	A mm	B mm	L mm
7.4	20	14.5	27.0	54.0
7.4	25	16.0	32.0	60.0
9.0	32	18.0	35.7	70.0
9.0	40	20.5	38.5	86.2
9.0	50	23.5	51.0	102.0
9.0	63	27.5	61.5	123.0
9.0	75	30.0	70.1	140.2
9.0	90	33.0	83.1	166.2
9.0	110	37.0	99.1	198.2
9.0	125	40.0	105.0	210.0

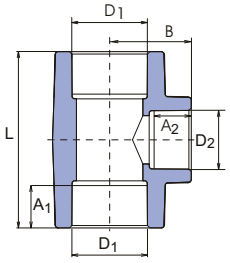




Reducing Tee

D_1 mm	D_2 mm	A_1 mm	A_2 mm	L mm	B mm
25	20	16.0	14.5	55.0	29.0
32	20	18.0	14.5	67.0	34.4
32	25	18.0	16.0	73.0	34.4
40	20	20.5	14.5	66.0	37.0
40	25	20.5	16.0	71.0	38.5
40	32	20.5	18.0	78.0	41.5
50	25	23.5	16.0	77.0	43.0
50	32	23.5	18.0	84.0	46.0
50	40	23.5	20.5	92.0	48.0
63	25	27.5	16.0	85.0	49.9
63	32	27.5	18.0	92.0	52.0
63	40	27.5	20.5	100.0	54.6
63	50	27.5	23.5	110.0	57.2
75	40	30.0	20.5	115.4	63.5
90	75	33.0	30.0	146.2	77.7

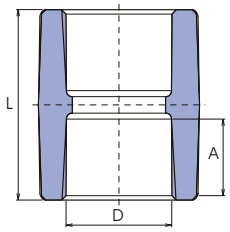




Socket

SDR	D mm	A mm	L mm
7.4	20	14.5	34.6
7.4	25	16.0	37.8
9.0	32	18.0	40.0
9.0	40	20.5	48.0
9.0	50	23.5	54.0
9.0	63	27.5	63.0
9.0	75	30.0	73.4
9.0	90	33.0	79.2
9.0	110	37.0	88.2
9.0	125	40.0	86.0

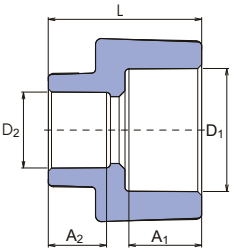




Reducer

SDR	D_1 mm	D_2 mm	A_1 mm	A_2 mm	L mm
7.4	25	20	16.0	14.5	34.0
7.4	32	20	18.0	14.5	39.0
9.0	32	25	18.0	16.0	39.5

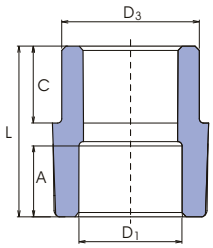




Reducer Internal / External

	D_3 mm	D_1 mm	A mm	L mm	C mm
	25	20	14.5	34.0	14.0
	32	20	14.5	40.0	16.0
	32	25	16.0	39.5	16.0
	40	20	14.5	36.3	18.5
	40	25	16.0	38.3	18.5
	40	32	18.0	41.3	18.5
	50	25	16.0	41.3	21.3
	50	32	18.0	43.3	22.0
	50	40	20.5	43.3	22.0
	63	25	16.0	45.3	25.3
	63	32	18.0	47.3	25.3
	63	40	20.5	44.3	25.3
	63	50	23.5	54.0	25.3
	75	40	20.5	64.5	28.9
	75	50	23.5	64.5	28.9
	75	63	27.5	71.5	28.9
	90	63	27.5	64.5	33.4
	90	75	30.0	78.0	33.4
	110	75	30.0	60.0	39.4
	110	90	33.0	87.0	39.4
	125	110	40.0	100.3	42.4

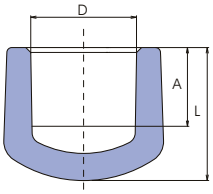




End Plug

SDR	D mm	A mm	B mm
7.4	20	14.5	24.5
7.4	25	16.0	28.0
9.0	32	18.0	33.0
9.0	40	20.5	34.0
9.0	50	23.5	49.0
9.0	63	27.5	46.5
9.0	75	30.0	60.0
9.0	90	33.0	69.0
9.0	110	37.0	79.0

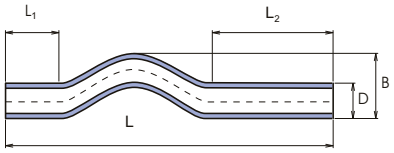




Crossover

SDR	D mm	B mm	L mm	L1 mm	L2 mm
7.4	20	47.0	400.0	80.0	180.0
7.4	25	52.0	400.0	80.0	120.0
9.0	32	65.0	390.0	64.0	106.0
9.0	40	65.0	390.0	64.0	106.0

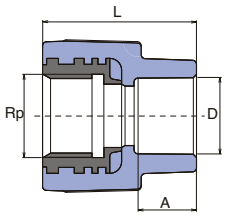




Reducing Sleeve with Metal Female Thread

SDR	D mm	Rp	A mm	L mm
7.4	20	1/2"	14.5	39.0
7.4	20	3/4"	14.5	42.0
7.4	25	1/2"	16.0	40.5
7.4	25	3/4"	16.0	47.0
9.0	32	3/4"	18.0	45.0
9.0	32	1"	18.0	57.0
9.0	40	5/4"	20.5	65.0
9.0	50	6/4"	23.5	69.0
9.0	63	2"	27.5	76.0
9.0	75	2½	28.0	65.5
9.0	90	3	32.0	90.0

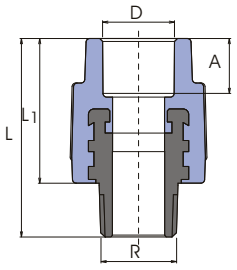




Reducing Sleeve with Metal Male Thread

SDR	D mm	Rp	A mm	L mm	L1 mm
7.4	20	1/2"	14.5	53.5	39.0
7.4	20	3/4"	14.5	58.0	40.0
7.4	25	1/2"	16.0	55.5	41.0
7.4	25	3/4"	16.0	59.0	41.0
9.0	32	3/4"	18.0	60.0	44.0
9.0	32	1"	18.0	62.5	46.0
9.0	32	1"	18.0	69.5	42.0
9.0	40	5/4"	20.5	80.0	48.0
9.0	50	6/4"	23.5	81.6	54.0
9.0	63	2"	27.5	101.0	63.5
9.0	75	2½"	30.0	110.0	69.7
9.0	90	3"	33.0	130.0	85.0

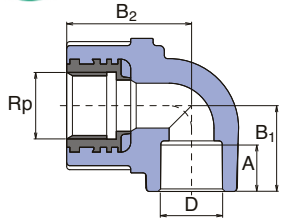




Elbow 90° with Metal Female Thread

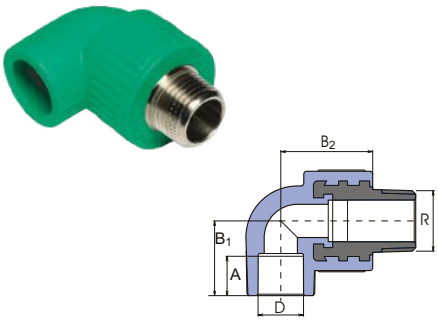
SDR	D mm	Rp	A mm	B1 mm	B2 mm
7.4	20	1/2"	14.5	27.0	34.5
7.4	20	3/4"	14.5	27.0	40.0
7.4	25	1/2"	16.0	31.0	36.0
7.4	25	3/4"	16.0	30.0	41.0
9.0	32	1"	18.0	36.0	51.0





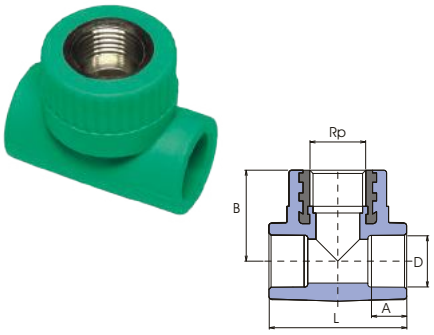
Elbow 90° with Metal Male Thread

SDR	D mm	Rp	A mm	B1 mm	B2 mm
7.4	20	1/2"	14.5	27.0	34.5
7.4	20	3/4"	14.5	27.0	39.0
7.4	25	1/2"	16.0	31.0	36.0
7.4	25	3/4"	16.0	30.0	39.0
9.0	32	1"	18.0	36.0	41.6



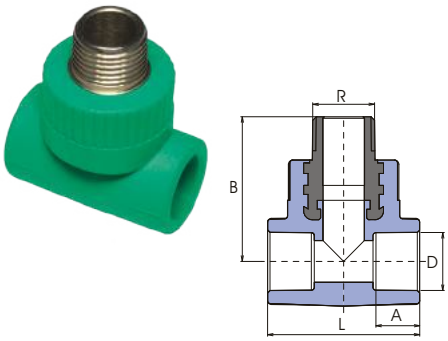
Tee with Metal Female Thread

SDR	D mm	Rp	A mm	L mm	B mm
7.4	20	1/2"	14.5	51.5	34.0
7.4	25	1/2"	16.0	80.0	40.0
7.4	25	3/4"	16.0	80.0	40.0
9.0	32	1/2"	18.0	71.0	40.5
9.0	32	3/4"	18.0	71.0	42.0
9.0	32	1"	18.0	80.0	55.0



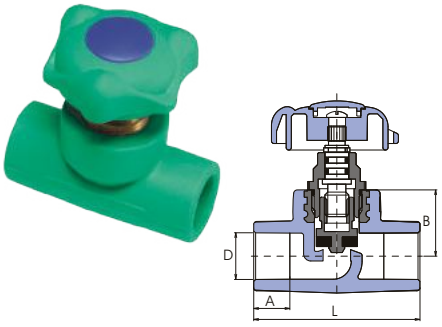
Tee with Metal Male Thread

SDR	D mm	Rp	A mm	L mm	B mm
7.4	20	1/2"	14.5	51.5	48.5
7.4	20	3/4"	14.5	60.0	50.0
7.4	25	1/2"	16.0	60.0	51.0
7.4	25	3/4"	16.0	65.0	52.5
9.0	32	3/4"	18.0	71.0	58.3
9.0	32	1"	18.0	80.0	60.0



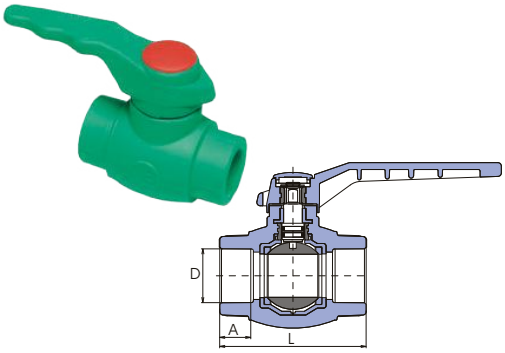
Straight – Way Valve

SDR	D mm	A mm	B mm	L mm
7.4	20	14.5	27.5	69.0
7.4	25	16.0	30.0	80.0
9.0	32	18.0	39.0	89.0
9.0	40	20.5	41.0	112.0
9.0	50	23.5	48.0	136.0
9.0	63	27.5	60.0	162.0



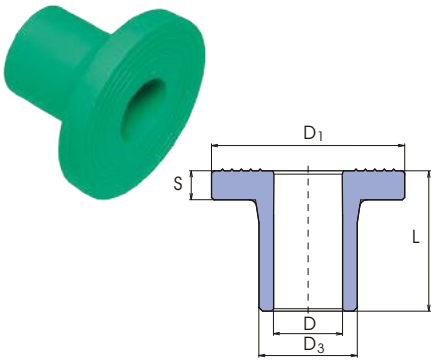
Ball Tap Plastic

SDR	D mm	A mm	L mm
7.4	20	14.5	65.0
7.4	25	16.0	71.0
9.0	32	18.0	85.0
9.0	40	20.5	100.0
9.0	50	23.5	115.0
9.0	63	27.5	134.0



Flange Adaptor

SDR	D_3 mm	L mm	D_1 mm	S mm	D mm
9.0	40	58.0	80.0	12.0	30.0
9.0	50	60.0	90.0	12.0	37.2
9.0	63	63.0	105.0	14.0	47.0
9.0	75	73.0	123.0	14.7	56.0
9.0	90	92.0	140.0	17.0	67.0
9.0	110	103.0	160.0	19.0	81.0

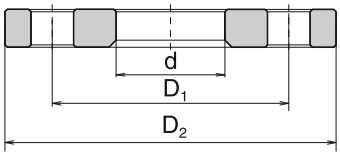


Flange

	D1 mm	D2 mm	d mm	Y
40 / DN32	100.0	140.0	46.0	4
50 / DN40	110.0	150.0	54.0	4
63 / DN50	125.0	165.0	66.0	4
75 / DN65	145.0	185.0	83.0	8
90 / DN80	160.0	200.0	94.0	8
110 / DN100	180.0	220.0	114.0	8
125 / DN150	210.0	250.0	166.0	8

Y – number of bore holes





Plastic Clip

D mm	20	25	32	40	50	63	75	90	110
				with tape	with tape	with tape	with tape	with tape	with tape

with tape



Plastic Double Clip

D mm	2×20	2×25



in grey colour

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LARGE DIAMETER FITTINGS (160mm - 250mm)

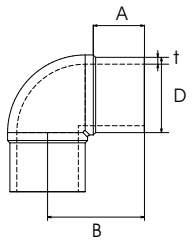
BUTT WELDING PROCESS



Elbow 90°

SDR	D mm	A mm	B mm	t mm
11.0	160	103	210	14.6
11.0	200	115	239	18.2
11.0	250	116	250	22.7

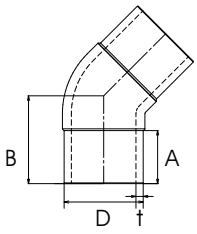




Elbow 45°

SDR	D mm	A mm	B mm	t mm
11.0	160	107	175	14.6
11.0	200	118	190	18.2
11.0	250	130	215	22.7

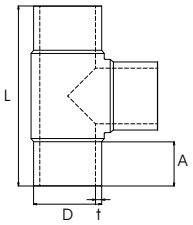




Tee

SDR	D mm	A mm	L mm	t mm
11.0	160	104	423	14.6
11.0	200	117	501	18.2
11.0	250	131	595	22.7

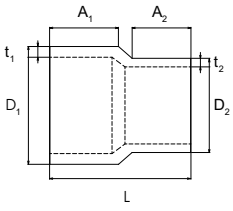




Reducer

SDR	D_1 mm	D_2 mm	t_1 mm	t_2 mm	A_1 mm	A_2 mm	L mm
11.0	160	110	14.6	-	106	92	226
11.0	160	125	14.6	-	105	93	216
11.0	200	160	18.2	14.6	117	100	240
11.0	250	160	22.7	14.6	60	56	162
11.0	250	200	22.7	18.2	128	116	274

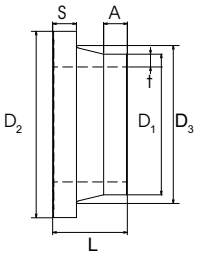




Flange Adaptor

SDR	D_1 mm	D_2 mm	D_3 mm	S mm	A mm	t mm	L mm
11.0	160	212	175	27	27	18.2	85
11.0	200	268	232	34	50	14.6	130
11.0	250	320	285	36	38	22.7	121





Working
the pipe

Butt welding process

We recommend using a special cutter for plastic pipes.

Butt welding is one way of coupling plastic piping systems and its components together. Butt welding is a process in which two pipe ends or a pipe end and a fitting are coupled by pressing the melted contact surfaces together. Butt welding can be performed only with welding equipment specifically designed for such purpose and only by appropriately qualified staff.

Butt welding is not suitable for coupling pipes and fittings of different size (the diameter and wall thickness must be the same) or different MFI value.

The following text describes the basic welding process.

Detailed welding equipment manuals including welding charts are provided by the manufacturer/supplier of the welding equipment.

In order to achieve more solid welds, we recommend using welding machines with hydraulic jaws.

The butt welding procedures described herein are based on DVS 2207.

Butt welding working instructions

Welding preparations

Workplace check

Make sure the workplace meets all the health and safety requirements and take note of dust and weather conditions. During the butt-welding process, the ambient temperature must not fall below 5 °C (use a mounting tent if needed). Similar measures must be taken in case of adverse weather conditions (rain, direct sunlight, etc.).

Welding equipment check

Check the technical condition of the welding machine (the surface and temperature of the heating plate, alignment of the fixed and movable jaws, milling cutter's functionality etc.).

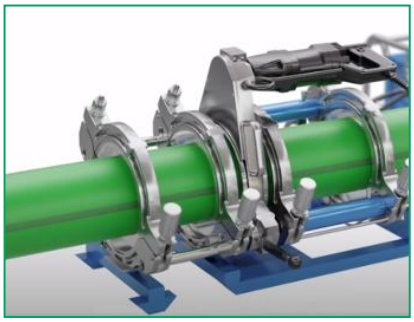
Materials check

Warning: Check the mutual weldability of the materials before starting the welding process. Ensure the materials you are planning to couple have the same temperature. Butt welding can only be performed with pipes with the same wall thickness (3 mm minimum).

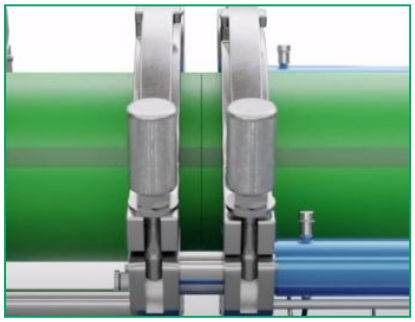
Preparing the materials

Once prepared and properly fastened, cut the pipe perpendicularly to the pipe axis with the tool designated for the purpose. Since the contact surface of the pipe must be clean, dry and free of any oils or fat, avoid using chainsaws with oil chain lubrication.

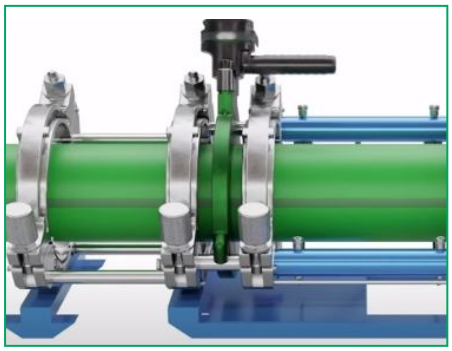
After completing the cut, always make sure to remove any burrs, sawdust or any other impurities created during the pipe cutting process. Check if the newly created contact surfaces fit well together; a careful inspection can reveal inadequate pipe ovality or tucked ends caused by the manufacturing process. Find out the passive resistance, plane the pipe surfaces as needed, test the fit once again, check for any gaps between the pipes, and finally clean the external surface of the pipes.



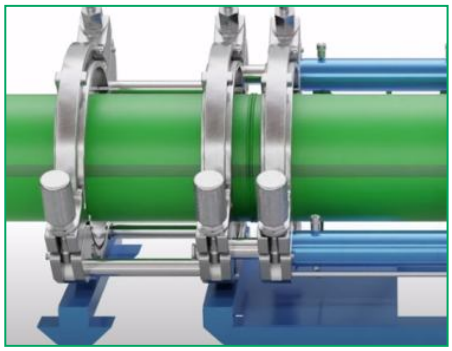
Planing the pipe ends



Properly prepared for welding



Heating with the heating plate



Cooling the weld seam

Welding phase

Truing phase
Push the contact surfaces against the heating plate until the two contact surfaces line up, which is indicated by the level of burr listed in the welding charts.

Heating phase
The contact surfaces should be heated with minimal contact pressure (see the welding charts for details). Heat up the contact surfaces until the welding zone reaches plastifcation (see the welding charts for details).

Positioning phase
Slide the contact surfaces away and remove the heating plate. Slide the pipes back as quickly as possible and push them together so their contact surfaces touch.

Coupling phase
When the contact surfaces are touching, raise the contact pressure until you reach full welding pressure.

Cooling phase
During the cooling phase keep the pipes under constant pressure (must be overseen the welder), until the time allotted for cooling runs out. After the weld is done, perform its visual inspection in accordance with the client's requirements.

Compatibility
In order to ensure proper welding compatibility, pipes and fittings meant for butt welding should not be combined with pipes or fittings produced by a different manufacturer.

Welding parameters
Included with the welding equipment.

Guideline values for polypropylene as specified in DVS 2207 - Part 11:
Hot body temperature: 200–220 °C

Welding pressure:
- Truing and coupling: 0.10 N/mm²
- Heating: 0.01 N/mm²

Welding time: divided according to the phases - see the chart below.

1	2	3	4	5	
	Truing	**Heating**	**Positioning**	**Coupling**	
	Hot body temperature: 210 +/- 10 °C				
Nominal wall thickness	Burr level on the hot body after truing period (min. values)	(heating p. = ≤ 0.01 N / mm²)	Positioning period (max.)	Coupling pressure rise period	Cooling under coupling pressure period (min. values) p = 0.10 N /mm² +/- 0.01
	(truing p = 0.10 N / mm²)				
[mm]	[mm]	[s]	[s]	[s]	[min]
up to 4.5	0.5	up to 135	5	6	6
4.5 - 7	0.5	135 - 175	5 - 6	6 - 7	6 - 12
7 - 12	1.0	175 - 245	6 - 7	7 - 11	12 - 20
12 - 19	1.0	245 - 330	7 - 9	11- 17	20 - 30
19 - 26	1.5	330 - 400	9 - 11	17 - 22	30 - 40
26 - 37	2.0	400 - 485	11 - 14	22 - 32	40 - 55
27 - 50	2.0	485 - 560	14 - 17	32 - 43	55 - 70

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